

Name : negation

Representation: $\sim$ (or) $\neg$

meaning: NOT

Truth table:

P	$\sim P$
T	F
F	T

Ex: P: London is in England

$\sim P$: London is not in England

Truth Table:

P	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Ex:

P: It is cold

q: It is raining

$P \vee q$ = It is cold OR it is raining

II Name: Conjunction

Representation: $\wedge$ (cap, meet)

meaning: AND

Truth table:

P	q	$P \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Ex: P: Ravi is studying very well

q: His father will buy a scooter

$P \wedge q$: Ravi is studying very well
AND His father will buy a scooter

IV Name: Conditional

Representation: $\rightarrow$, $\Rightarrow$ (Implication)

meaning: If-then

Truth Table:

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

ex:

P: Today is Tuesday

q: Tomorrow is Wednesday

$P \rightarrow q$: If Today is Tuesday
then tomorrow is Wednesday

Note: $P \rightarrow q$

P is called "Antecedent" or
"Hypothesis"

III Name: Disjunction

Representation: $\vee$ (cup, Join)

meaning: OR

q is called "consequent" or "conclusion".

V Name: Bi-conditional

Representation: $\leftrightarrow, \iff$ (equivalence)

Meaning: If & only if

Truth table:

P	q	$P \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Ex:

P: He swims

q: The water is warm

$P \leftrightarrow q$: He swims if & only if the water is warm.

Truth table construction.

① $(P \wedge (P \rightarrow q))$

P	q	$(P \rightarrow q)$	$(P \wedge (P \rightarrow q))$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	F

$$2) (P \rightarrow (q \rightarrow r))$$

P	q	r	$(q \rightarrow r)$	$(P \rightarrow (q \rightarrow r))$
T	T	T	T	T
T	T	F	F	F
T	F	T	T	T
T	F	F	T	T
F	T	T	T	T
F	T	F	F	T
F	F	T	T	T
F	F	F	T	T

$$3) [q \wedge (P \rightarrow q)] \rightarrow P$$

$$4) (P \rightarrow q) \vee (\sim P \rightarrow q)$$

$$5) (\sim(P \vee q) \vee r) \rightarrow \sim p$$

Tautology:

A compound Proposition is always true, irrespective of the truth values of the Proposition that occurs in it is called "Tautology".

Contradiction:

A compound proposition is always false, irrespective of the truth values of the Proposition that occurs in it is called "Contradiction".

Contingency:

A compound proposition is neither tautology nor contradiction is called "Contingency".

① $[q \wedge (p \rightarrow q)] \rightarrow p$

P	q	$p \rightarrow q$	$(q \wedge (p \rightarrow q))$	$[q \wedge (p \rightarrow q)] \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	F

② $(p \rightarrow q) \vee (\sim p \rightarrow q)$

P	q	$p \rightarrow q$	$\sim p$	$(\sim p \rightarrow q)$	$(p \rightarrow q) \vee (\sim p \rightarrow q)$
T	T	T	F	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	F	T	T	F	T

$$\textcircled{3} (\sim(p \vee q) \vee r) \rightarrow \sim p$$

~~True~~

p	q	r	$p \vee q$	$\sim(p \vee q)$	$(\sim(p \vee q) \vee r)$	$\sim p$	$(\sim(p \vee q) \vee r) \rightarrow \sim p$
T	T	T	T	F	T	F	F
T	T	F	T	F	F	F	T
T	F	T	T	F	T	F	F
T	F	F	T	F	F	F	T
F	T	T	T	F	T	T	T
F	T	F	T	F	F	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

① show that $(p \vee q) \vee \sim p$ is a Tautology.

p	q	$\sim p$	$p \vee q$	$(p \vee q) \vee \sim p$
T	T	F	T	T
T	F	F	T	T
F	T	T	T	T
F	F	T	F	T

$\therefore$ The given statement formula is Tautology.

② Prove that, $(p \wedge (p \Rightarrow q)) \rightarrow q$ is a Tautology

p	q	$p \Rightarrow q$	$(p \wedge (p \Rightarrow q))$ A	$A \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

$\therefore$ The given statement formula is Tautology.

3. Prove that $[(P \wedge \sim Q) \rightarrow R] \rightarrow [P \rightarrow (Q \vee R)]$ is a Tautology using Truth Table.

P	Q	R	$\sim Q$	$(P \wedge \sim Q)$	$(Q \vee R)$	$\frac{[(P \wedge \sim Q) \rightarrow R]}{A}$	$\frac{[P \rightarrow (Q \vee R)]}{B}$	$A \rightarrow B$
T	T	T	F	F	T	T	T	T
T	T	F	F	F	T	T	T	T
T	F	T	T	T	T	T	T	T
T	F	F	T	T	F	F	F	T
F	T	T	F	F	T	T	T	T
F	T	F	F	F	T	T	T	T
F	F	T	T	F	T	T	T	T
F	F	F	T	F	F	T	T	T

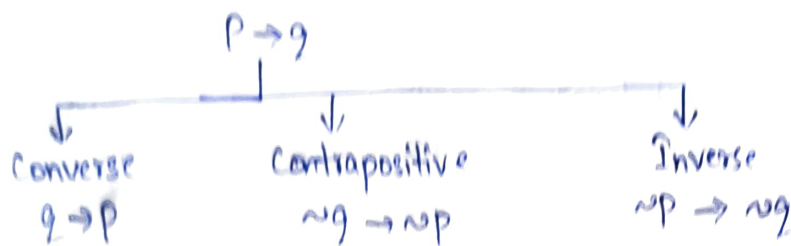
$\therefore$ The given statement formula is Tautology

(4) Show that, $Q \vee (P \wedge \sim Q) \vee (\sim(P \wedge \sim Q))$ is tautology

* 5) Establish that $\sim\{[(P \vee Q) \rightarrow r] \wedge (\sim P)\} \rightarrow (Q \rightarrow r)$ is a Contra.

P	Q	R	$\sim P$	$P \vee Q$	$Q \rightarrow r$	$\frac{[(P \vee Q) \rightarrow r]}{A}$	$\frac{A \wedge (\sim P)}{B}$	$[B \rightarrow (Q \rightarrow r)]$	$\sim[B \rightarrow (Q \rightarrow r)]$
T	T	T	F	T	T	T	F	T	F
T	T	F	F	T	F	F	F	T	F
T	F	T	F	T	T	T	F	T	F
T	F	F	F	T	T	F	F	T	F
F	T	T	T	T	T	T	T	T	F
F	T	F	T	T	F	T	T	T	F
F	F	T	T	F	T	T	T	T	F
F	F	F	T	F	F	T	T	T	F

Converse, Contrapositive, Inverse



1) If $\triangle ABC$ is a right angled triangle then $(AC)^2 = (AB)^2 + (BC)^2$

Sol let us indicate the statement as follows.

P : $\triangle ABC$ is a right angled triangle

Q : $(AC)^2 = (AB)^2 + (BC)^2$

Converse:

If $(AC)^2 = (AB)^2 + (BC)^2$ then $\triangle ABC$ is right angled triangle

Contrapositive:

If $(AC)^2 \neq (AB)^2 + (BC)^2$ then $\triangle ABC$ is NOT a right angled triangle.

Inverse:

If $\triangle ABC$ is NOT a right angled triangle then $(AC)^2 \neq (AB)^2 + (BC)^2$

2) what are the contra-positive, the inverse, the converse of the following conditional statement?

Sol If you work hard, then you will be rewarded.

Let's indicate the statements as following

P : you work hard

R : you will be rewarded.

Converse:

If you will be rewarded then you work hard.

Contrapositive:

If you will not be rewarded then you don't work hard.

Inverse:

you will be rewarded.

Equivalent formulas:

1) $P \vee P \equiv P$ (Idempotent law)
 $P \wedge P \equiv P$

2) $P \vee Q \equiv Q \vee P$ (Commutative law)
 $P \wedge Q \equiv Q \wedge P$

3) $P \vee (Q \vee R) \equiv (P \vee Q) \vee R$ (Associative law)
 $P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$

4) $P \wedge T \equiv P$ (Identity law)
 $P \vee F \equiv P$

5) $P \vee T \equiv T$ (Domination law)
 $P \wedge F \equiv F$

*6) $\sim(P \vee Q) \equiv \sim P \wedge \sim Q$
 $\sim(P \wedge Q) \equiv \sim P \vee \sim Q$ (De Morgan's law)

*7) $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$
 $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$ (Distributive law)

8) $\sim(\sim P) \equiv P$ (Double negation law)

9) $P \vee \sim P \equiv T$ (Negation law)
 $P \wedge \sim P \equiv F$

10) $P \vee (P \wedge Q) \equiv P$ (Absorption law)
 $P \wedge (P \vee Q) \equiv P$

Note:

*1) $P \rightarrow Q \equiv \sim P \vee Q$
(or)
 $\sim Q \rightarrow \sim P$

2) $P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$
 $(P \wedge Q) \vee (\sim P \wedge \sim Q)$

Logically Equivalent:

If A, B are two statement formulae are said to be logically equivalent, then $A \leftrightarrow B$ is Tautology.

1) show that $\sim(P \vee (\sim P \wedge Q))$ and $(\sim P \wedge \sim Q)$ are logically equivalent.

$\sim(P \vee (\sim P \wedge Q))$ and $(\sim P \wedge \sim Q)$

$$\sim(P \vee (\sim P \wedge Q)) \Leftrightarrow \sim P \wedge \sim(\sim P \wedge Q) \text{ (DeMorgan's law)}$$

$$\Leftrightarrow \sim P \wedge (P \vee \sim Q) \text{ (Double Neg. Law)}$$

$$\Leftrightarrow (\sim P \wedge P) \vee (\sim P \wedge \sim Q) \text{ (Distributive law)}$$

$$\Leftrightarrow F \vee (\sim P \wedge \sim Q) \text{ (Negation law)}$$

$$\Leftrightarrow \sim P \wedge \sim Q \text{ (Identity law)}$$

$\therefore \sim(P \vee (\sim P \wedge Q))$ and $(\sim P \wedge \sim Q)$ are logically equivalent.

P	Q	$\sim P$	$\sim Q$	$(\sim P \wedge Q)$	$P \vee (\sim P \wedge Q)$	$\underbrace{\sim(P \vee (\sim P \wedge Q))}_A$	$\underbrace{\sim P \wedge \sim Q}_B$	$A \leftrightarrow B$
T	T	F	F	F	F	T	F	F
T	F	F	T	F	F	T	F	F
F	T	T	F	T	F	T	F	F
F	F	T	T	F	F	T	F	F

2) show that $\sim(P \rightarrow Q)$ and $P \wedge \sim Q$

P	Q	$\sim Q$	$P \rightarrow Q$	$\underbrace{\sim(P \rightarrow Q)}_A$	$\underbrace{P \wedge \sim Q}_B$	$A \leftrightarrow B$
T	T	F	F	T	F	T
T	F	T	T	F	T	T
F	T	F	T	F	F	T
F	F	T	T	F	F	T

without using truth-table:

1. To eliminate $\rightarrow, \leftrightarrow$ by using suitable expressions
2. To eliminate $\sim$ before the paranthesis ' $()$ ' by using Demorgan's law.
3. By using Demorgans laws, Distributive laws repeatedly

★ 1) show that, $(\sim p \wedge (\sim q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \Leftrightarrow r$

$$(\sim p \wedge (\sim q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \Leftrightarrow [(\sim p \wedge \sim q) \wedge r] \vee [(q \vee p) \wedge r]$$

($\because$ Associative & Distributive laws)

$$\Leftrightarrow [(\sim p \wedge \sim q) \vee (q \vee p)] \wedge r \text{ (Distributive law)}$$

$$\Leftrightarrow [\sim(p \vee q) \vee (p \vee q)] \wedge r \text{ (Demorgans \& Commutative law)}$$

$$\Leftrightarrow T \wedge r \text{ (negation law)}$$

$$\Leftrightarrow r \text{ (Identity law)}$$

$$\therefore (\sim p \wedge (\sim q \wedge r)) \vee (q \wedge r) \vee (p \wedge r) \Leftrightarrow r$$

2) Show that, $(p \wedge q) \rightarrow (p \vee q)$ is a Tautology

$$(p \wedge q) \rightarrow (p \vee q) \Leftrightarrow \sim[p \wedge q] \vee (p \vee q)$$

$$(\because p \rightarrow q \Leftrightarrow \sim p \vee q)$$

$$\Leftrightarrow [\sim p \vee \sim q] \vee (p \vee q)$$

($\because$ Demorgans law)

$$\Leftrightarrow [\sim p \vee p] \vee [\sim q \vee q] \text{ ($\because$ commutative \& Associative laws)}$$

$$\Leftrightarrow T \vee T \text{ (negation law)}$$

$$\Leftrightarrow T \text{ (Idempotent law)}$$

Duality Law:

If A, B are two statement formulas are said to be Dual to each other, one can be obtained from the other by replacing ' $\wedge$ ' by ' $\vee$ ' (or) ' $\vee$ ' by ' $\wedge$ ' and ' T ' by ' F ' (or) ' F ' by ' T '.

Ex:

1. Write the duality of the following expression

$$[(P \wedge Q) \rightarrow r]$$

$$[(P \vee Q) \rightarrow r]$$

2. Duality of $[P \rightarrow Q] \wedge T$ is $[P \rightarrow Q] \vee F$.

Well formed formula:

A statement formula consisting by a string containing the connectivities, Symbols, paranthesis.

Rules of well formed formula:

1. A statement variable standing along is a well formed formula.
2. If A is a well formed formula, then $\neg A$ is also a well formed formula.
3. If A, B are two statement formulas, then $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$, $(A \leftrightarrow B)$ are also well formed formulas.

Ex: $P \wedge Q$ $\times$

$$(P \wedge Q) \rightarrow r \quad \times$$

$$\neg P \vee Q \quad \times$$

$$\neg (P \wedge Q) \quad \checkmark$$

$$((P \vee Q) \rightarrow r) \quad \checkmark$$

Normal form:

Normal forms

Disjunctive
Normal form (DNF)

↓
Principle Disjunctive
Normal form (PDNF)
(T)

$$(A) \vee (A)$$

Conjunctive
Normal form (CNF)

↓
Principle conjunctive
Normal form (PCNF)
(F)

$$(A) \wedge (A)$$

PDNF:

A statement formula which is equivalent to given formula consisting of disjunction of "min-terms" only is called "Principle Disjunction Normal forms" (PDNF) (or) sum of product canonical form.

1) $\wedge \rightarrow$ Product

2) $\vee \rightarrow$ sum

3) min-terms

$\frac{P, Q}{(2)}$

$\frac{P, Q, \neg}{(3)}$

$P \wedge Q$

$P \wedge Q \wedge \neg$

$\neg P \wedge Q$

$\neg P \wedge Q \wedge \neg$

$P \wedge \neg Q$

$P \wedge \neg Q \wedge \neg$

$\neg P \wedge \neg Q$

$P \wedge \neg Q \wedge \neg$

$\neg P \wedge \neg Q \wedge \neg$

$\neg P \wedge \neg Q \wedge \neg$

$P \wedge \neg Q \wedge \neg$

$\neg P \wedge \neg Q \wedge \neg$

3) $(\neg) \vee (\neg) \rightarrow$ sum of product canonical form.

$(\vee) \wedge (\vee) \rightarrow$ product of sum canonical form.

① Find the PDNF of the following

$$(\neg P \vee Q) \Leftrightarrow (\neg P \wedge T) \vee (Q \wedge T) \text{ [Identity law]}$$

$$\Leftrightarrow (\neg P \wedge (Q \vee \neg Q)) \vee (Q \wedge (P \wedge \neg P)) \text{ [Negation law]}$$

$$\Leftrightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (Q \wedge P) \vee (Q \wedge \neg P) \text{ [Distributive law]}$$

$$\Leftrightarrow (\neg P \wedge Q) \wedge (\neg P \wedge \neg Q) \vee (P \wedge Q) \text{ [Commutative law \& Idempotent law]}$$

P Q $\neg P$ $\neg P \vee Q$

T T F T

T F F F

F T T T

F F T T

min terms are

$(P \wedge Q), (\neg P \wedge Q), (\neg P \wedge \neg Q)$

PDNF is

$(P \wedge Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q)$

2) Find the PDNF of $(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$

$$(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r) \Leftrightarrow [(p \wedge q) \wedge T] \vee [(\neg p \wedge r) \wedge T] \vee [(q \wedge r) \wedge T]$$

$$\Leftrightarrow [(p \wedge q) \wedge (r \vee \neg r)] \vee [(\neg p \wedge r) \wedge (q \vee \neg q)] \vee [(q \wedge r) \wedge (p \vee \neg p)]$$

$$\Leftrightarrow (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge r \wedge q) \vee (\neg p \wedge r \wedge \neg q) \vee (q \wedge r \wedge p) \vee (q \wedge r \wedge \neg p)$$

$$\Leftrightarrow (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (\neg p \wedge r \wedge q) \vee (\neg p \wedge r \wedge \neg q) \vee (p \wedge r \wedge q) \vee (p \wedge r \wedge \neg q)$$

$\therefore$ It is the required PDNF

Truth table method.

P	q	r	$\neg p$	$(p \wedge q)$	$(\neg p \wedge r)$	$(q \wedge r)$	$(p \wedge q) \vee (\neg p \wedge r) \vee (q \wedge r)$	$A \vee (q \wedge r)$
T	T	T	F	T	F	T	T	T
T	T	F	F	T	F	F	T	T
T	F	T	F	F	F	F	F	F
T	F	F	F	F	F	F	F	F
F	T	T	T	F	T	T	T	T
F	T	F	T	F	F	F	F	F
F	F	T	T	F	F	F	F	F
F	F	F	T	F	F	F	F	F

3) Find the PDNF of $P \rightarrow [C \rightarrow Q] \wedge \neg (\neg P \vee \neg Q)$

$$\Leftrightarrow P \rightarrow [(\neg P \vee Q) \wedge (Q \wedge P)] \quad (\text{De Morgan's law})$$

$$\Leftrightarrow P \rightarrow [(\neg P \wedge Q \wedge P) \vee (Q \wedge Q \wedge P)] \quad (\text{Distributive law})$$

$$\Leftrightarrow P \rightarrow [(F \wedge Q) \vee (Q \wedge P)] \quad (\text{Idempotent \& Negation law})$$

$$\Leftrightarrow P \rightarrow [F \vee (Q \wedge P)] \quad (\text{domination law})$$

$$\Leftrightarrow P \rightarrow [P \wedge Q] \quad (\text{Identity law})$$

$$\Leftrightarrow \neg P \vee (P \wedge Q) \quad (P \rightarrow Q \Rightarrow \neg P \vee Q)$$

$$\Leftrightarrow [\neg P \wedge T] \vee [P \wedge Q] \quad (\text{Identity law})$$

$$\Leftrightarrow [\neg P \wedge (P \vee \neg Q)] \vee [P \wedge Q] \quad (\text{Inverse law})$$

$$\Leftrightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (P \wedge Q) \quad (\text{D.L})$$

4) Find the PDNF of $\sim[p \vee (\sim p \wedge \sim q \wedge r)]$

$$\sim[p \vee (\sim p \wedge \sim q \wedge r)] \Leftrightarrow \sim p \wedge \sim(\sim p \wedge \sim q \wedge r)$$

$$\sim[p \vee (\sim p \wedge \sim q \wedge r)] \Leftrightarrow \sim p \wedge \sim[\sim p \wedge \sim q \wedge r]$$

$$\Leftrightarrow \sim p \wedge (p \vee q \vee \sim r)$$

$$\Leftrightarrow (\sim p \wedge p) \vee (\sim p \wedge q) \vee (\sim p \wedge \sim r)$$

$$\Leftrightarrow F \vee (\sim p \wedge q) \vee (\sim p \wedge \sim r)$$

$$\Leftrightarrow (\sim p \wedge q) \vee (\sim p \wedge \sim r)$$

$$\Leftrightarrow [(\sim p \wedge q) \wedge T] \vee [(\sim p \wedge \sim r) \wedge T]$$

$$\Leftrightarrow [(\sim p \wedge q) \wedge (r \vee \sim r)] \vee [(\sim p \wedge \sim r) \wedge (q \vee \sim q)]$$

$$\Leftrightarrow (\sim p \wedge q \wedge r) \vee (\sim p \wedge q \wedge \sim r) \vee (\sim p \wedge \sim r \wedge q) \vee (\sim p \wedge \sim r \wedge \sim q)$$

PCNF (Principle Conjunctive normal form):

A statement formula which is equivalent to given formula consisting of conjunction of "Maxterms" only is called "PCNF" (or) Product of sum canonical form.

Maxterms:

P, q

$p \vee q$

$\sim p \vee q$

$p \vee \sim q$

$\sim p \vee \sim q$

P, q, r

$p \vee q \vee r$

$\sim p \vee q \vee r$

$p \vee \sim q \vee r$

$p \vee q \vee \sim r$

$\sim p \vee \sim q \vee r$

$\sim p \vee q \vee \sim r$

$p \vee \sim q \vee \sim r$

$\sim p \vee \sim q \vee \sim r$

1) Find the PCNF of the following statement.

$$\sim(p \vee q) \Leftrightarrow \sim p \wedge \sim q$$

$$\Leftrightarrow (\sim p \vee F) \wedge (\sim q \vee F)$$

$$\Leftrightarrow (\sim p \vee (q \wedge \sim q)) \wedge (\sim q \vee (p \wedge \sim p))$$

$$\Leftrightarrow (\sim p \vee q) \wedge (\sim p \vee \sim q) \wedge (\sim q \vee p) \wedge (\sim q \vee \sim p)$$

$$\Leftrightarrow (\sim p \vee q) \wedge (\sim p \vee \sim q) \wedge (\sim q \vee p)$$

	q	$p \vee q$	$\sim(p \vee q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

max terms,

$$(\sim p \vee \sim q), (\sim p \vee q), (p \vee \sim q)$$

The required PCNF is,

$$(\sim p \vee \sim q) \wedge (\sim p \vee q) \wedge (p \vee \sim q)$$

2) Find PCNF of $\sim(p \rightarrow q)$

p	q	$p \rightarrow q$	$\sim(p \rightarrow q)$
T	T	T	F
T	F	F	T
F	T	T	F
F	F	T	F

Max terms;

$$(\sim p \vee \sim q), (p \vee \sim q), (p \vee q)$$

The required PCNF is,

$$(\sim p \vee \sim q) \wedge (p \vee \sim q) \wedge (p \vee q)$$

without truth table:

$$\sim(p \rightarrow q) \Leftrightarrow \sim(\sim p \vee q)$$

$$\Leftrightarrow p \wedge \sim q$$

$$\Leftrightarrow (p \vee F) \wedge (\sim q \vee F)$$

$$p \vee (q \wedge \sim q)$$

Imp

$$P \rightarrow Q \equiv \sim P \vee Q$$

$$\sim P \rightarrow Q = P \vee Q$$

Theory of Inference

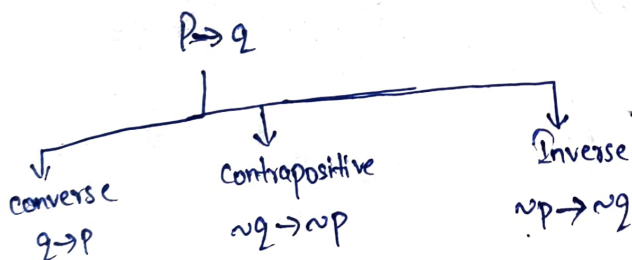
- ① Premises
(or)
Axioms
(or)
Hypothesis
∴ conclusion

② $P_1, P_2, P_3, \dots, P_n$
∴ c

③ $P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n$
∴ c

④ $[P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n] \rightarrow C$

Conjunction of premises conditional conclusion.



Rules of Inference:

① $\frac{P \wedge Q}{\therefore P}$ or $\frac{P \wedge Q}{\therefore Q}$ → By rule of simplification.

② $\frac{P}{\therefore P \vee Q}$ (or) $\frac{Q}{\therefore P \vee Q}$ → By rule of addition

③ $\frac{P}{Q}$
∴ $P \wedge Q$ → By rule of conjunction.

$$\begin{array}{l} P \\ P \rightarrow q \\ \hline \therefore q \end{array} \quad (or) \quad \begin{array}{l} P \rightarrow q \\ P \\ \hline \therefore q \end{array} \quad \rightarrow \text{By rule of Modus ponens}$$

$$\textcircled{5} \quad \begin{array}{l} P \vee q \\ \neg p \\ \hline \therefore q \end{array} \quad \rightarrow \text{By rule of Modus Tollens}$$

$$\textcircled{6} \quad \begin{array}{l} P \rightarrow q \\ q \rightarrow r \\ \hline \therefore P \rightarrow r \end{array} \quad \rightarrow \text{Transitive Rule}$$

$$\textcircled{7} \quad \begin{array}{l} P \rightarrow q \\ \neg q \\ \hline \therefore \neg P \end{array} \quad \rightarrow \text{Disjunction sylloism (contrapositive)}$$

$$\textcircled{8} \quad \begin{array}{l} (P \rightarrow q) \wedge (r \rightarrow s) \\ (p \vee r) \\ \hline \therefore (q \vee s) \end{array} \quad \rightarrow \text{Disjunctive Dilemma}$$

$$\textcircled{9} \quad \begin{array}{l} (P \rightarrow q) \wedge (r \rightarrow s) \\ (\neg q \vee \neg s) \\ \hline \therefore (\neg P \vee \neg r) \end{array} \quad \rightarrow \text{Constructive Dilemma}$$

Method 1:

1) Demonstrate that r is a valid inference from the premises $P \rightarrow q$, $q \rightarrow r$ and P

Given Premises are

$$P \rightarrow q, q \rightarrow r, P$$

Conclusion is

$$r$$

$\{1\}$ (1) $P \rightarrow Q$ Rule P

$\{2\}$ (2) $Q \rightarrow r$ Rule P

$\{1,2\}$ (3) $P \rightarrow r$ Transitive Rule

$\{4\}$ (4) P Rule P

$\{1,2,4\}$ (5) r Modus Ponens

2) show that $(r \vee s)$ follows logically from the premises
 $(c \vee d), (c \vee d) \rightarrow \neg h, \neg h \rightarrow (a \wedge b), (a \wedge b) \rightarrow (r \vee s)$

sol The given premises are

$(c \vee d), (c \vee d) \rightarrow \neg h, \neg h \rightarrow (a \wedge b), (a \wedge b) \rightarrow (r \vee s)$

The conclusion is

$(r \vee s)$

$\{1\}$ (1) $c \vee d$ Rule P

$\{2\}$ (2) $(c \vee d) \rightarrow \neg h$ Rule P

$\{1,2\}$ (3) $\neg h$ Modus Ponens

$\{4\}$ (4) $\neg h \rightarrow (a \wedge b)$ Rule P

$\{1,2,4\}$ (5) $a \wedge b$ Modus Ponens

$\{6\}$ (6) $(a \wedge b) \rightarrow (r \vee s)$ Rule P

$\{1,2,4,6\}$ (7) $r \vee s$ Modus Ponens

3) show that $s \vee x$ is tautologically implied by
 $(p \vee q) \wedge (p \rightarrow x) \wedge (q \rightarrow s)$

The given premises are
 $(p \vee q), (p \rightarrow r), (q \rightarrow s)$

The conclusion is
 $s \vee r$

{1}	(1)	$p \vee q$	Rule P
{1}	(2)	$\neg p \rightarrow q$	Rule T
{3}	(3)	$q \rightarrow s$	Rule P
{1,3}	(4)	$\neg p \rightarrow s$	Transitive Rule
{1,3}	(5)	$\neg s \rightarrow p$	contrapositive
{6}	(6)	$p \rightarrow r$	Rule P
{1,3,6}	(7)	$\neg s \rightarrow r$	Transitive Rule
{1,3,6}	(8)	$s \vee r$	Rule T

(Q) Show that $\neg(p \wedge q)$ follows from $(\neg p \wedge \neg q)$

sol we prove this by indirect method of proof

(or) Law of contradiction

Hence we consider additional premises and which leads to contradiction

{1}	(1)	$\neg(\neg p \wedge \neg q)$	p (assumed)
{1}	(2)	$(p \wedge q)$	T
{1,2}	(3)	p	T (simplification)
{4}	(4)	$(\neg p \wedge \neg q)$	P
{4}	(5)	$\neg p$	T (simplification)
{1,2,4}	(6)	$p \wedge \neg p$	

which is contradiction

Hence

$\neg(P \wedge Q)$ follows $\neg P \vee \neg Q$

4) Show that $R \wedge (P \vee Q)$ is a valid conclusion from the Premises $P \vee Q$, $Q \rightarrow R$, $P \rightarrow M$ and $\sim M$

sol The given premises are

$P \vee Q$, $Q \rightarrow R$, $P \rightarrow M$, and $\sim M$

The conclusion is

$R \wedge (P \vee Q)$

{1} (1) $P \vee Q$ Rule P

{1} (2) $\sim P \rightarrow Q$ Rule T

{3} (3) $Q \rightarrow R$ Rule P

{1,3} (4) $\sim P \rightarrow R$ T.R

{1,3} (5) $\sim R \rightarrow P$ Contradictive

{6} (6) $P \rightarrow M$ Rule P

{1,3,6} (7) $\sim R \rightarrow M$ T.R

{1,3,6} (8) $\sim M \rightarrow R$ Contrapositive

{9} (9) $\sim M$ Rule P

{1,3,6,9} (10) R M.P

{1,3,6,9} (11) $R \wedge (P \vee Q)$ Conjunction.

Method-2:

1) ~~Test~~ Test the validity of the following argument: "Soniya is watching T.V. If Soniya is watching T.V, then she is not studying. If she is not studying, then her father will not buy a scooty. Therefore, soniya's father will not buy a scooty."

Sol: Let us indicate the statements as follows

P: Soniya is watching T.V

Q: Soniya is studying

R: Soniya's father will buy a scooty

The given premises are

$$P, P \rightarrow \sim Q, \sim Q \rightarrow \sim R$$

The conclusion is

$$\sim R$$

$$\{1\} (1) \quad P \quad \text{Rule P}$$

$$\{2\} (2) \quad P \rightarrow \sim Q \quad \text{Rule P}$$

$$\{1, 2\} (3) \quad \sim Q \quad \text{M.P}$$

$$\{1, 2\} (4) \quad \sim Q \rightarrow \sim R \quad \text{Rule P}$$

$$\{1, 2, 4\} (5) \quad \sim R \quad \text{M.P}$$

2) "If there was a Ball game, then travelling was difficult. If they arrived on time, then travelling was not difficult. They arrived on time. Therefore, there was no ball game." show that these statements constitute a valid argument.

sol Let us indicate the statements as follows

P: There was a ball game

Q: Travelling was difficult

R: They arrived on time

The given premises are

$$P \rightarrow Q, R \rightarrow \neg Q, R$$

The conclusion is

$$\neg P$$

$$\{1\} \quad (1) \quad P \rightarrow Q \quad \text{Rule P}$$

$$\{2\} \quad (2) \quad R \rightarrow \neg Q \quad \text{Rule P}$$

$$\{2\} \quad (3) \quad Q \rightarrow \neg R \quad \text{Contrapositive}$$

$$\{1, 2\} \quad (4) \quad P \rightarrow \neg R \quad \text{T.R}$$

$$\{1, 2\} \quad (5) \quad R \rightarrow \neg P \quad \text{contrapositive}$$

$$\{6\} \quad (6) \quad R \quad \text{Rule P}$$

$$\{1, 2, 6\} \quad (7) \quad \neg P \quad \text{M.P}$$

imp
3) If Clifton does not live in France then he does not speak French. Clifton does not drive a Datsun. If Clifton live in France then he rides a ~~bicycle~~ ~~bicycle~~ ~~either~~ bicycle. Either Clifton speaks French or he drives a Datsun. Hence, Clifton rides a bicycle.

Let us indicate the statements as follows.

P : Clifton live in France

Q : ~~he~~ Clifton speaks French

R : Clifton drives a Datsun

S : Clifton rides a bicycle

The given premises are

$$\sim P \rightarrow \sim Q, \sim R, P \rightarrow S, Q \vee R$$

The conclusion is

S

$\{1\}$ (1) $Q \vee R$ Rule P

$\{1\}$ (2) $\sim Q \rightarrow R$ Rule T

$\{1\}$ (3) $\sim R \rightarrow Q$ contrapositive

$\{4\}$ (4) $\sim R$ Rule P

$\{1, 4\}$ (5) Q M.P

$\{6\}$ (6) $\sim P \rightarrow \sim Q$ Rule P

$\{6\}$ (7) $Q \rightarrow P$ contrapositive

$\{1, 4, 6\}$ (8) P Modus Ponens

$\{9\}$ (9) $P \rightarrow S$ Rule P

$\{1, 4, 6, 9\}$ (10) S Modus Ponens.

Method 3: ($\rightarrow$)

(9m) Rule CP: (Conditional Proof):

We can derive S from R set of premises, then we obtain the conclusion is $R \rightarrow S$.

Note: The conclusion is of the form $P \rightarrow Q$ then we take P as a additional premises or assumed premises

1) Show that ~~$r \rightarrow s$~~ $r \rightarrow s$ can be derived from the

Premises $[P \rightarrow (Q \rightarrow S)], \neg \vee P, Q$

sol: The given premises are

$[P \rightarrow (Q \rightarrow S)], \neg \vee P, Q$

The conclusion is

$r \rightarrow s$

{1} (1) r Rule P (Additional premises)

{2} (2) $\neg \vee P$ Rule P

{2} (3) $r \rightarrow P$ Rule T

{1,2} (4) P Modus Ponens

{5} (5) $[P \rightarrow (Q \rightarrow S)]$ Rule P

{1,2,5} (6) $Q \rightarrow S$ Modus Ponens

{7} (7) Q Rule P

{1,2,5,7} (8) S M.P

{1,2,5,7} (9) $r \rightarrow s$ Rule C.P

2) By using the method of derivation show that the following statements constitute a valid argument: "If A works hard, then either B or C will enjoy. If B enjoy, then A will not work hard. If B enjoy, then C will not. Therefore, if A works hard, D will not enjoy"

Let us indicate the statements as follows

P : A works hard

Q : B will enjoy

R : C will enjoy

S : D will enjoy

The given premises are

$[P \rightarrow (Q \vee R)], Q \rightarrow \neg P, S \rightarrow \neg R$

The conclusion is

$P \rightarrow \neg S$

{1}	(1)	P	Rule P (Additional Premises)
{2}	(2)	$Q \rightarrow \neg P$	Rule P
{2}	(3)	$P \rightarrow \neg Q$	contrapositive
{1, 2}	(4)	$\neg Q$	M.P
	(5)	$P \rightarrow (Q \vee R)$	Rule

{1} (1) P Rule P (Additional Premises)

{2} (2) $P \rightarrow (Q \vee R)$ Rule P

{1,2} (3) $Q \vee R$ M.P

{1,2} (4) $\neg Q \rightarrow R$ Contrapositive

{5} (5) $Q \rightarrow \neg P$ Rule P

{5} (6) $P \rightarrow \neg Q$ Contrapositive

{1,2,5} (7) $R \rightarrow P$ T.R

{8} (8) $S \rightarrow \neg R$ Rule P

{8} (9) $R \rightarrow \neg S$ Contrapositive

{1,2,5,8} (10) $P \rightarrow \neg S$ T.R

Method 4: Indirect method of Proof

1) Show that $\neg(P \wedge Q)$ follows from $\neg p$ and $\neg q$

The given premises are
 $\neg p$ and $\neg q$

The conclusion is
 $\neg(P \wedge Q)$

{1} (1) $\neg[\neg(P \wedge Q)]$ Rule P (A.P)

{1} (2) $P \wedge Q$ Double negation law

{1} (3) P Simplification

{2} (4) Q Simplification

{5} (5) $\neg p \wedge \neg q$ Rule P

{5} (6) $\neg p$ simplification

{5} (7) $\neg q$ simplification

{1,5} (8) $p \wedge \neg p$ conjunction

$$p \wedge \neg p \equiv F$$

$\therefore$ our Assumption is wrong

$\neg(p \wedge \neg q)$ follows from $\neg p \wedge \neg q$

2) show that $p \rightarrow q, q \rightarrow r, \neg(p \wedge r), p \vee r \Leftrightarrow r$ using indirect method of proof

The given premises are

$$p \rightarrow q, q \rightarrow r, \neg(p \wedge r), p \vee r$$

The conclusion is

$$r$$

{1} (1) $\neg r$ Rule P (A-P)

{2} (2) $p \rightarrow q$ Rule P

{3} (3) $q \rightarrow r$ Rule P

{2,3} (4) $p \rightarrow r$ Transitive Rule

{5} (5) $\neg(p \wedge r)$ Rule P

{5} (6) $\neg p \vee \neg r$ D.L

{5} (7) $p \rightarrow \neg r$ Rule T ($p \rightarrow q \equiv \neg p \vee q$)

{8} (8) $p \vee r$ Rule P

{8} (9) $\neg p \rightarrow r$ Rule T

$\{1, 2, 3\}$ (10) $\vee$ p M.T

$\{1, 2, 3, 8\}$ (11) $\neg$ M.P

$\{1, 2, 3, 8\}$ (12) $\neg \wedge \neg$ conjunction

$$\neg \wedge \neg \neg \equiv F$$

$\therefore$ Our assumption is wrong

Method 3 :-

consistency of premises

1. show that the premises $[a \rightarrow (b \rightarrow c)]$, $[d \rightarrow (b \wedge \neg c)]$, and are inconsistency.

The given premises are $[a \rightarrow (b \rightarrow c)]$, $[d \rightarrow (b \wedge \neg c)]$, and

$\{1\}$ (1) and Rule P

$\{1\}$ (2) a simplification

$\{1\}$ (3) d simplification

$\{1\}$ (4) $[a \rightarrow (b \rightarrow c)]$ Rule P

~~$\{1, 4\}$ (5) $b \rightarrow c$ Modus Ponens~~

$\{1, 4\}$ (6) $\neg b \vee c$ Rule T

$\{1, 4\}$ (7) $\neg [b \wedge \neg c]$ Demorgan's Law

$\{8\}$ (8) $[d \rightarrow (b \wedge \neg c)]$ Rule P

$\{1, 8\}$ (9) $b \wedge \neg c$ Modus Ponens

(10) $(b \wedge \neg c) \wedge \neg (b \wedge \neg c)$ conjunction

$$\therefore (b \wedge \neg c) \wedge \neg (b \wedge \neg c) \equiv F$$

$\therefore$ Given premises are Inconsistent

Q.2) show that following set of premises is inconsistent: If the contract is valid, then John is liable for penalty. If John is liable for penalty, he will go bankrupt. If the bank will loan him money, he will not go bankrupt. As a matter of fact, the contract is valid and the bank will loan him money".

Sol Let us indicate the statements as follows

- P : The contract is valid
 q : John is liable for penalty
 r : John will go bankrupt
 s : The bank will loan him money

The given premises are

$$P \rightarrow q, q \rightarrow r, s \rightarrow \neg r, P \wedge s$$

- | | | | |
|--------------|------|--|-----------------|
| {1} | (1) | $P \rightarrow q$ | Rule P |
| {2} | (2) | $q \rightarrow r$ | Rule P |
| {1, 2} | (3) | $P \rightarrow r$ | Transitive Rule |
| {4} | (4) | $s \rightarrow \neg r$
$s \rightarrow \neg r$ | Rule P |
| {4} | (5) | $r \rightarrow \neg s$ | Contrapositive |
| {1, 2, 4} | (6) | $P \rightarrow \neg s$ | T.R |
| {1, 2, 4} | (7) | $\neg P \vee \neg s$ | Rule T |
| {1, 2, 4} | (8) | $\neg [P \wedge s]$ | Demorgan's law |
| {9} | (9) | $P \wedge s$ | Rule P |
| {1, 2, 4, 9} | (10) | $(P \wedge s) \wedge \neg (P \wedge s)$ | conjunction |

$$(P \wedge S) \wedge \neg(P \wedge S) \equiv F$$

$\therefore$ our premises are inconsistent.

3) show that the following premises are inconsistent:

- i) If ~~Jack~~ ^{Jack} misses many classes because of illness, then he fail high school.
- ii) If Jack fails high school, then he is uneducated
- iii) If Jack reads a lot of books, then he is not uneducated.
- iv) Jack misses many classes because of illness and reads a lot of books

Let us ~~consider~~ indicate the statements as follows

P: Jack misses many classes because of illness

Q: Jack fail high school

R: Jack is uneducated

S: Jack reads lots of books

The given premises are

$$P \rightarrow Q, Q \rightarrow R, S \rightarrow \neg R, P \wedge S$$

$$\{1\} \quad (1) \quad P \rightarrow Q \quad \text{Rule P}$$

$$\{2\} \quad (2) \quad Q \rightarrow R \quad \text{Rule P}$$

$$\{1, 2\} \quad (3) \quad P \rightarrow R \quad \text{Transitive Rule}$$

$$\{4\} \quad (4) \quad S \rightarrow \neg R \quad \text{Rule P}$$

$\{4\}$ (5) $r \rightarrow \sim s$

contrapositive

$\{1, 2, 4\}$ (6) $p \rightarrow \sim s$

Transitive Rule

~~$\{1, 2, 4\}$ (7) $\sim p \vee \sim s$~~

~~Rule T~~

$\{1, 2, 4\}$ (7) $\sim p \vee \sim s$

Rule T

$\{1, 2, 4\}$ (8) $\sim [p \wedge s]$

Demorgan's law

$\{9\}$ (9) $p \wedge s$

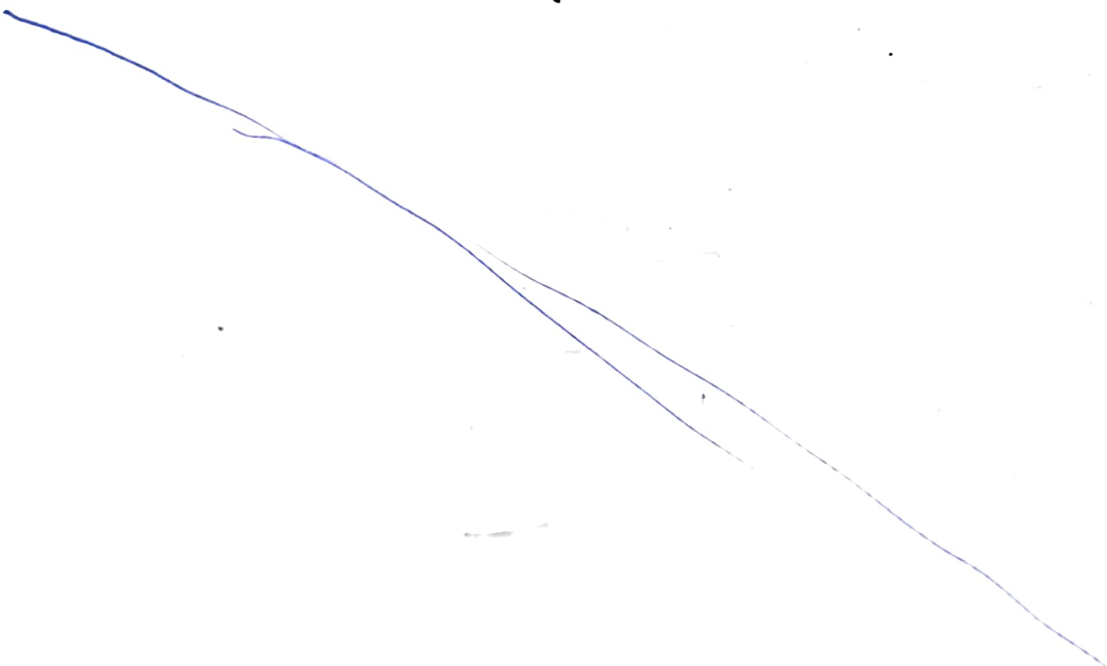
Rule P

$\{1, 2, 4, 9\}$ (10) $(p \wedge s) \wedge \sim (p \wedge s)$

Conjunction

$$(p \wedge s) \wedge \sim (p \wedge s) \equiv F$$

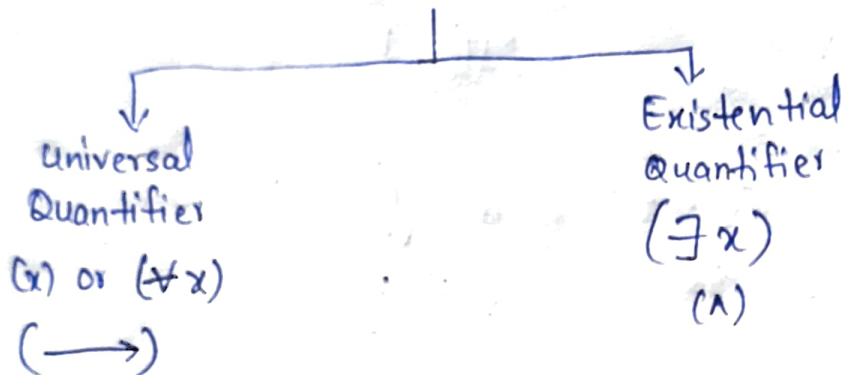
$\therefore$ Our premises are inconsistent



Quantifiers :

^{2m} The phrase which indicates the quantity of the statement is called quantifier

Quantifiers



Every x

Everything x

all x

everyone x

Some x
there exist x
something x

Rules of Quantifiers:

Rule us (universal specification)

$$(\forall x) A(x) \Rightarrow A(y)$$

All → one

Rule uG (universal Generalization)

$$A(y) \Rightarrow \forall x A(x)$$

one → All

Rule ES (Existential specification)

$$(\exists x) A(x) \Rightarrow A(y)$$

Some → one

Rule EG (Existential Generalization)

$$A(y) \Rightarrow (\exists x) A(x)$$

one → some

Method 1:

1) show that $(x)[P(x) \rightarrow Q(x)] \wedge (x)[Q(x) \rightarrow R(x)] \Rightarrow (x)[P(x) \rightarrow R(x)]$

The given premises are $(x)[P(x) \rightarrow Q(x)]$, $(x)[Q(x) \rightarrow R(x)]$

The conclusion is $(x)[P(x) \rightarrow R(x)]$

{1} (1) $(x)[P(x) \rightarrow Q(x)]$ Rule P

{1} (2) $P(y) \rightarrow Q(y)$ Rule US

{3} (3) $(x)[Q(x) \rightarrow R(x)]$ Rule P

{3} (4) $Q(y) \rightarrow R(y)$ Rule US

{1,3} (5) $P(y) \rightarrow R(y)$ Transitive Rule

{1,3} (6) $(x)[P(x) \rightarrow R(x)]$ Rule UG

2) show that $(\exists x)M(x)$ follows logically from the premises

$(x)[H(x) \rightarrow M(x)]$ and $(\exists x)H(x)$

The given premises are $(x)[H(x) \rightarrow M(x)]$, $(\exists x)H(x)$

The conclusion is $(\exists x)M(x)$

{1} (1) $(x)[H(x) \rightarrow M(x)]$ Rule P

{1} (2) $H(y) \rightarrow M(y)$ Rule US

{3} (3) $(\exists x)H(x)$ Rule P

{3} (4) $H(y)$ Rule ES

{1,3} (5) $M(y)$ Modus Ponens

{1,3,5} (6) $(\exists x)M(x)$ Rule EG

Method 2:

- 1) verify the validity of the following argument: "All men are mortal. Socrates is a man. Therefore, Socrates is mortal."

The predicate form is

$M(x)$: x is a man

$H(x)$: x is mortal

s : Socrates

The premises are

$$(x)[M(x) \rightarrow H(x)], M(s)$$

The conclusion is

$H(s)$

$$\{1\} (1) (x)[M(x) \rightarrow H(x)] \quad \text{Rule P}$$

$$\{1\} (2) M(s) \rightarrow H(s) \quad \text{Rule U.S.}$$

$$\{3\} (3) M(s) \quad \text{Rule P}$$

$$\{1, 3\} (4) H(s) \quad \text{M.P.}$$

- 2) Verify the validity of the following Argument:

"Every living thing is a plant (or) An animal. Joe's goldfish is alive and it is not a plant. All animals have hearts. Therefore Joe goldfish has a heart"

sol The predicate form is

$P(x)$: x is a plant

$A(x)$: x is an animal

$H(x)$: x has heart

F : Joe's gold fish

The premises are

$$(x)[P(x) \vee A(x)], \sim P(f), (x)[A(x) \rightarrow H(x)]$$

The conclusion is

$$H(f)$$

$$\{1\} \quad (1) \quad (x)[P(x) \vee A(x)] \quad \text{Rule P}$$

$$\{1\} \quad (2) \quad P(f) \vee A(f) \quad \text{Rule U.S}$$

$$\{1\} \quad (3) \quad \sim P(f) \rightarrow A(f) \quad \text{Rule T}$$

$$\{4\} \quad (4) \quad \sim P(f) \quad \text{Rule P}$$

$$\{1, 4\} \quad (5) \quad A(f) \quad \text{M.P}$$

$$\{6\} \quad (6) \quad (x)[A(x) \rightarrow H(x)] \quad \text{Rule P}$$

$$\{6\} \quad (7) \quad A(f) \rightarrow H(f) \quad \text{Rule U.S}$$

$$\{1, 4, 6\} \quad (8) \quad H(f) \quad \text{Modus Ponens}$$

3) Establish the validity of the following argument:
"All integers are Rational numbers. Some integers are Powers of two. Therefore, some rational numbers are Powers of 2".

The predicate form is

$I(x)$: x is an integer

$R(x)$: x is a rational number

$P(x)$: x is a powers of two

The premises are

$$(x)[I(x) \rightarrow R(x)], (\exists x)[I(x) \wedge P(x)]$$

The conclusion is

$$(\exists x)[R(x) \wedge P(x)]$$

$$\{1\} \quad (1) \quad (x)[I(x) \rightarrow R(x)] \quad \text{Rule P}$$

$$\{1\} \quad (2) \quad I(a) \rightarrow R(a) \quad \text{Rule US}$$

$$\{3\} \quad (3) \quad (\exists x)[I(x) \wedge P(x)] \quad \text{Rule P}$$

$$\{3\} \quad (4) \quad I(a) \wedge P(a) \quad \text{Rule ES}$$

$$\{4\} \quad (5) \quad I(a) \quad \text{simplification}$$

$$\{4\} \quad (6) \quad P(a) \quad \text{simplification}$$

$$\{1, 4\} \quad (7) \quad R(a) \quad \text{Modus Ponens}$$

$$\{ \quad \} \quad (8) \quad R(a) \wedge P(a) \quad \text{Conjunction}$$

$$(9) \quad (\exists x)[R(x) \wedge P(x)] \quad \text{Rule EG}$$

Unit-2

1. Principle of inclusion and exclusion:
2. Equivalence relation
3. Hasse diagram
4. Transitive ~~close~~ closer
5. Bijective function
6. Inverse function and composition of functions

Principle of inclusion and exclusion:

- 1) $|A \cup B| = |A| + |B| - |A \cap B|$
- 2) $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$
- 3) $|A \cup B \cup C \cup D| = |A| + |B| + |C| + |D| - |A \cap B| - |B \cap C| - |C \cap A| - |D \cap A| - |A \cap C| - |B \cap D| + |A \cap B \cap C| + |B \cap C \cap D| + |C \cap D \cap A| + |B \cap A \cap D| - |A \cap B \cap C \cap D|$

Q1) How many positive integers not exceeding 2000 are divisible by 7 or 11

Sol Given that

$$U = \{1, 2, 3, \dots, 2000\}$$

$$|U| = 2000$$

Let

A denotes the number of positive integers which are divisible by 7.

Let B denotes the number of positive integers which are divisible by 11.

Let $A \cap B$ denotes the set of positive integers which are divisible by 7 and 11 ($7 \times 11 = 77$)

$$|A| = \frac{2000}{7} = 285$$

$$|B| = \frac{2000}{7} = 181$$

$$|A \cap B| = \frac{2000}{77} = 25$$

By the principle of inclusion and exclusion we have

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 285 + 181 - 25$$

$$= 441$$

2) How many integers from 1 to 100 are multiple of 2 or 3

imp 3) Determine the number of positive integers n , where $1 \leq n \leq 2000$ and n is not divisible by 2, 3 or 5 but is divisible by 7

sol Let $U = \{1, 2, 3, \dots, 2000\}$

$$|U| = 2000$$

Let A denotes the number of positive integers which are divisible by 2.

Let B denotes "

are divisible by 3

Let C " " " " " "

are divisible by 5

Let $A \cap B$ denotes the set of integers which are divisible by 2 and 3
 (2 x 3 = 6)
 Let $B \cap C$ " " " " " " " " by 3 and 5
 (3 x 5 = 15)
 Let $C \cap A$ " " " " " " " " by 5 and 2
 (5 x 2 = 10)
 Let $A \cap B \cap C$ denotes " " " " " " " " divisible by
 2, 3 and 5 (2 x 3 x 5 = 30)

$$|A| = \frac{2000}{2} = 1000$$

$$|B| = \frac{2000}{3} = 666$$

$$|C| = \frac{2000}{5} = 400$$

$$|A \cap B| = \frac{2000}{6} = 333$$

$$|B \cap C| = \frac{2000}{15} = 133$$

$$|C \cap A| = \frac{2000}{10} = 200$$

$$|A \cap B \cap C| = \frac{2000}{30} = 66$$

By the principle of inclusion and exclusion we have

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C| \\ &= 1000 + 666 + 400 - 333 - 133 - 200 + 66 \\ &= 1466 \end{aligned}$$

$$\begin{aligned} |(A \cup B \cup C)^c| &= |U| - |A \cup B \cup C| \\ &= 2000 - 1466 \\ &= 534 \end{aligned}$$

The number of integers which are not divisible by 2, 3 or 5 but divisible by 7 $\Rightarrow \frac{534}{7} = 76$

4) How many integers between 1 and 1000 are not divisible by 2, 3, 5, 7

sol

Let $u = \{1, 2, 3, \dots, 1000\}$

$|v| = 1000$

Let A denotes the number of positive integers which are divisible by 2.

Let B denotes " " " " " divisible by 3.

и с и и и и и и и 5.

" D " " " " " " 7.

" A n B " " " " " " by both 2 and 3' ($2 \times 3 = 6$)

" BNC " " " " " " " " 3 and 5 ($3 \times 5 = 15$)

" CAP " " " " " " " 5 and 7 ($5 \times 7 = 35$)

"DNA" = " " " " " " 7 and 2 ($7 \times 2 = 14$)

" AND " " " " " " " 2.5 (2x5=10)

11 BOD, 4, 11, 11, 11, 11, 11, 3 and 7 (2x2)

10. $A \cap B \cap C$ " " " " " " " 2.03.5 (2x3x5 = 30)

[illegible]

11 CNDA 11 11 11 11 11 5, 7 & 2 (5x7x2=70)

" BRAND " " " " " " 3, 2 & 7 (3x2x7=42)

11. $ANBNCND$ denotes the number of positive integers which

are divisible by 2, 3, 5 and 7 ($2 \times 3 \times 5 \times 7 = 210$)

$$|A| = \frac{1000}{2} = 500$$

$$|B| = \frac{1000}{3} = 333$$

$$|C| = \frac{1000}{5} = 200$$

$$|D| = \frac{1000}{7} = 142$$

$$|A \cap B| = \frac{1000}{6} = 166$$

$$|B \cap C| = \frac{1000}{15} = 66$$

$$|C \cap D| = \frac{1000}{35} = 28$$

$$|D \cap A| = \frac{1000}{14} = 71$$

$$|A \cap C| = \frac{1000}{10} = 100$$

$$|B \cap D| = \frac{1000}{21} = 47$$

$$|A \cap B \cap C| = \frac{1000}{30} = 33$$

$$|B \cap C \cap D| = \frac{1000}{105} = 9$$

$$|C \cap D \cap A| = \frac{1000}{70} = 14$$

$$|B \cap A \cap D| = \frac{1000}{42} = 23$$

$$|A \cap B \cap C \cap D| = \frac{1000}{210} = 4$$

By Principle of inclusion and exclusion we have

$$\begin{aligned} |A \cup B \cup C \cup D| &= |A| + |B| + |C| + |D| - |A \cap B| - |B \cap C| - |C \cap D| - |D \cap A| - |A \cap C| \\ &\quad - |B \cap D| + |A \cap B \cap C| + |B \cap C \cap D| + |C \cap D \cap A| + |B \cap A \cap D| \\ &\quad - |A \cap B \cap C \cap D| \end{aligned}$$

$$= 500 + 333 + 200 + 142 - 166 - 66 - 28 - 71 - 100 - 47 + 33 + 9 + 14 + 23 - 4$$

$$= 772$$

$$|(A \cup B \cup C \cup D)'| = |U| - |A \cup B \cup C \cup D|$$

$$= 1000 - 772$$

$$= 228$$

5) Use principle of inclusion and exclusion, find the total number of integers between 1 and 1000 which are neither perfect squares nor perfect cubes

Sol

$$\text{Let } U = \{1, 2, 3, \dots, 1000\}$$

$$|U| = 1000$$

Let A denotes the set of integers which are perfect squares

$$A = \{1^2, 2^2, 3^2, \dots, 31^2\}$$

$$|A| = 31$$

Let B denotes the set of positive integers which are perfect Cubes

$$B = \{1^3, 2^3, 3^3, \dots, 10^3\}$$

$$|B| = 10$$

Let $A \cap B$ denotes the set of integers which are perfect squares and perfect cubes

$$A \cap B = \{1^6, 2^6, 3^6\}$$

$$|A \cap B| = 3$$

~~By~~ ~~pr~~

By principle of inclusion and exclusion we have

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 31 + 10 - 3$$

$$= 38$$

$$|(A \cup B)^c| = |U| - |A \cup B|$$

$$= 1000 - 38$$

6) Out of 250 candidates who failed in an examination it was revealed that 128 failed in mathematics, 87 in Physics and 134 in aggregate. 31 failed in mathematics and in physics, 54 failed in the aggregate and in mathematics, 30 failed in the aggregate and in physics. Find how many candidates failed

- i) In all three subjects
- ii) In mathematic but not in physics
- iii) In the aggregate but not in mathematics

sol Let U denotes the total number of candidates.

$$|U| = 250 \quad (\text{or}) \quad |A \cup B \cup C| = 250$$

Let A denotes the number of candidates failed in Mathematics

Let B " " " " " " " " physics

Let C " " " " " " " " Aggregate

Let $A \cap B$ " " " " " " " " failed both

Mathematics and physics.

Let $B \cap C$ " " " " " " " " failed in both physics

and Aggregate.

Let $C \cap A$ " " " " " " " " Mathematics

and aggregates.

$$|A| = 128$$

$$|B| = 87$$

$$|C| = 134$$

$$|A \cap B| = 31$$

$$|B \cap C| = 30$$

$$|C \cap A| = 54$$

By the principle of inclusion and exclusion we have

$$(i) |A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$

$$250 = 128 + 87 + 134 - 31 - 30 - 54 + |A \cap B \cap C|$$

$$250 = 234 + |A \cap B \cap C|$$

$$(ii) |A \cap B \cap C| = 250 - 234$$

$$|A \cap B \cap C| = 16$$

$$(iii) |A \cap B'| = |A| - |A \cap B|$$

$$= 128 - 31$$

$$= 97$$

$$(iv) |C \cap A'| = |C| - |C \cap A|$$

$$= 134 - 54$$

$$= 80$$

7) A total of 1232 students have taken a course in Spanish. 879 have taken a course in French and 114 have taken a course in Russian. Further 100 have taken course in both Spanish and French, 25 have taken course in both Spanish and Russian, and 12 have taken course in both French and Russian. If 2092 students have taken atleast one of Spanish, French, and Russian courses, how many

students have taken a course in all three languages.

sol:

Let U denotes the total number of 2092 students

$$|U| = 2092$$

Let A denotes the no. of students taken Spanish course

" B " " " " " " " French "

" C " " " " " " " Russian "

" $A \cap B$ " " " " " " " taken both Spanish and French

" $B \cap C$ " " " " " " " French and Russian

" $C \cap A$ " " " " " " " Spanish and Russian

$$|A| = 1232$$

$$|B| = 879$$

$$|C| = 114$$

$$|A \cap B| = 100$$

$$|B \cap C| = 12$$

$$|C \cap A| = 25$$

By principle of inclusion and exclusion we have

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$

$$2092 = 1232 + 879 + 114 - 100 - 12 - 25 + |A \cap B \cap C|$$

$$2092 = 2088$$

$$|A \cap B \cap C| = 2092 - 2088$$

$$|A \cap B \cap C| = 4$$

Sol: Given that,

$$U = \{1, 2, 3, \dots, 100\}$$

$$|U| = 100$$

Let A be the set of integers which are multiples of 2

$$|A| = \frac{100}{2} = 50$$

Let B be the set of integers which are multiple of 3

$$|B| = \frac{100}{3} = 33$$

Let $A \cap B$ denote the set of positive integers which are ~~and~~ multiples of both 2 and 3 ($2 \times 3 = 6$)

$$|A \cap B| = \frac{100}{6} = 16$$

By the principle of Inclusion and Exclusion, we have

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 50 + 33 - 16$$

$$= 67$$

Relations:

Let X be a non-empty set and R is a relation defined on X .

1) Reflexive Relation: Let X be a non-empty set and R is a Relation defined on X is said to be reflexive Relation then
 $a R a \quad \forall a \in X$

ex: The relation ' $\leq$ ' is satisfies reflexive relation

2) Symmetric Relation: Let X be a non-empty set and R is a relation defined on X is said to be a symmetric relation if aRb then bRa
ex: The relation '=' satisfies symmetric relation

3) Anti symmetric Relation: Let X be a non empty set and R is a relation defined on X is said to be Anti symmetric relation if aRb & bRa then $a=b$.

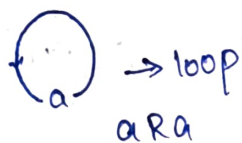
ex: The relation ' $\leq$ ' satisfies anti symmetric relation.

4) Transitive Relation: Let X be a non empty set and R is a relation defined on X is said to be Transitive relation if aRb & bRc then aRc .

ex: The relation '=' satisfies Transitive relation

Graphical Representation of Relation:

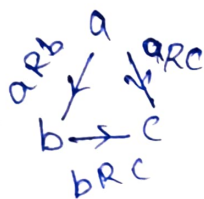
1. R.R:



2. S.R:



3. T.R:



1) Let $X = \{1, 2, 3, 4\}$ and the Relation $R = \{(1,1) (1,3) (2,1) (3,1) (3,3) (4,3)\}$ find the Relation matrix and draw the digraph

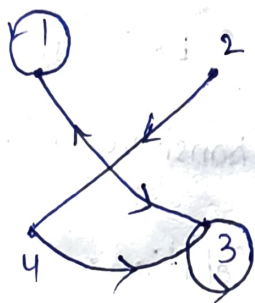
Sol: Given that

$$X = \{1, 2, 3, 4\}$$

$$R = \{(1,1) (1,3) (2,1) (3,1) (3,3) (4,3)\}$$

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

digraph:



2) Let $X = \{1, 2, 3, 4\}$ and Relation matrix is

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \text{ and draw the digraph}$$

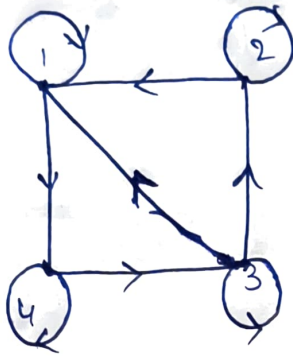
Sol: Given that

$$X = \{1, 2, 3, 4\}$$

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

$$R = \{(1,1) (1,4) (2,1) (2,2) (3,1) (3,2) (3,3) (4,3) (4,4)\}$$

digraph:



	compatibility Relation	Equivalence relation	Partial ordering relation
R.O.R	✓	✓	✓
S.O.R	✓	✓	✓
A.S.O.R			✓
T.O.R		✓	✓

~~Exm)~~
Hasse diagram:-

The pictorial representation of partial ordering relation is called Hasse diagram

i) Let $P = \{1, 2, 3, 4\}$ draw the Hasse diagram for the relation ' $\leq$ '

sol:- Step 1:-

Given that

$P = \{1, 2, 3, 4\}$ and the relation

is ' $\leq$ '

Step 2 :-

$$(P, \leq) = \{(1,1) (1,2) (1,3) (1,4) (2,2) (2,3) (2,4) (3,3) (3,4) (4,4)\}$$

Step 3 :-



$$\text{infimum} = 1$$

$$\text{supremum} = 4$$

$$\text{No. of edges} = 3$$

2) Draw the Hasse diagram for a divisibility relation R on A where

i) $A = \{1, 2, 3, 6, 12, 24\}$

ii) $A = \{1, 2, 4, 8, 16, 32, 64\}$

sol

i) Step 1 :-

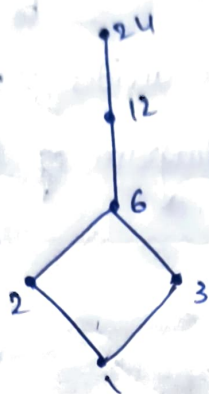
Given that

$A = \{1, 2, 3, 6, 12, 24\}$ and the relation is ~~divisibility~~ divisibility

Step 2 :-

$$(A, |) = \{(1,1) (2,2) (3,3) (6,6) (12,12) (24,24) (1,2) (1,3) (1,6) (1,12) (1,24) (2,6) (2,12) (2,24) (3,6) (3,12) (3,24) (6,12) (6,24) (12,24)\}$$

Step 3:



infimum = 1

Supremum = 24

No. of edges = 6

ii) Step 1:

Given that,

$A = \{1, 2, 4, 8, 16, 32, 64\}$ and the relation is divisibility

Step 2:-

$$(A, |) = \{(1,1) (2,2) (4,4) (8,8) (16,16) (32,32) (64,64) (1,2) (1,4) (1,8) (1,16) (1,32) (1,64) (2,4) (2,8) (2,16) (2,32) (2,64) (4,8) (4,16) (4,32) (4,64) (8,16) (8,32) (8,64) (16,32) (16,64) (32,64)\}$$

Step 3:-



3) If $X = \{2, 3, 6, 12, 24\}$ and if $\leq$ be the partial order, defined by $x \leq y$ if x divides y . Determine the number of edges in Hasse diagram of $(X, \leq)$

sol:

Step 1:

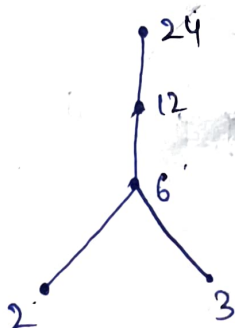
Given that

$X = \{2, 3, 6, 12, 24\}$ and the relation $\leq$

Step 2:

$$(X, \leq) = \{(2, 2) (3, 3) (6, 6) (12, 12) (24, 24) (2, 6) (2, 12) (2, 24) (3, 6) (3, 12) (3, 24) (6, 12) (6, 24) (12, 24)\}$$

Step 3:



Infimum = 2, 3

Supremum = 24

No. of edges = 4

★ u) Draw the Hasse diagram for the divisibility on the set $\{1, 2, 3, 6, 12, 24, 36, 48, 96\}$

sol: Step 1:

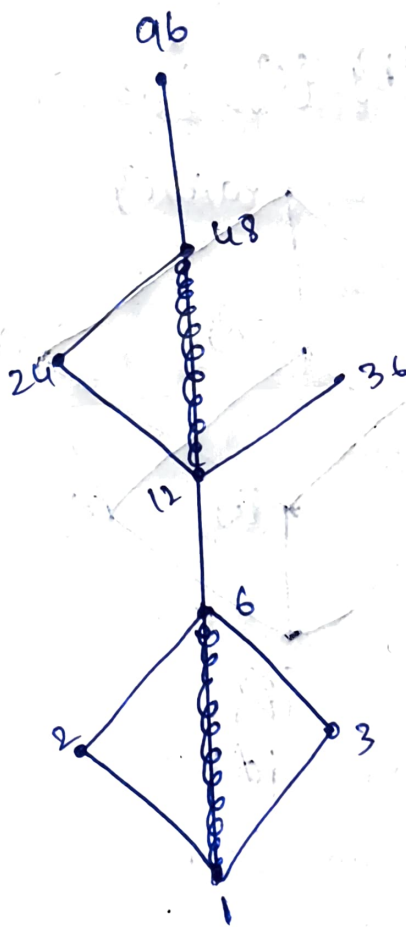
Given that

$X = \{1, 2, 3, 6, 12, 24, 36, 48, 96\}$ and the relation is divisibility.

Step 2:

$$(X, 1) = \{(1,1) (1,2) (3,3) (6,6) (12,12) (24,24) (36,36) (48,48) (96,96) (1,2) (1,3) (1,6) (1,12) (1,24) (1,36) (1,48) (1,96) (2,6) (2,12) (2,24) (2,36) (2,48) (2,96) (3,6) (3,12) (3,24) (3,36) (3,48) (3,96) (6,12) (6,24) (6,36) (6,48) (6,96) (12,24) (12,36) (12,48) (12,96) (24,48) (24,96) (48,96)\}$$

Step 3:



infimum = 1

Supremum = 96, 36

No of edges = 9

5) Let $B = \{a, b, c\}$ and $A = P(B)$ p be power set of B . Draw the Hasse diagram for ' $\subseteq$ ' and Poset A .

Sol: Step 1:-

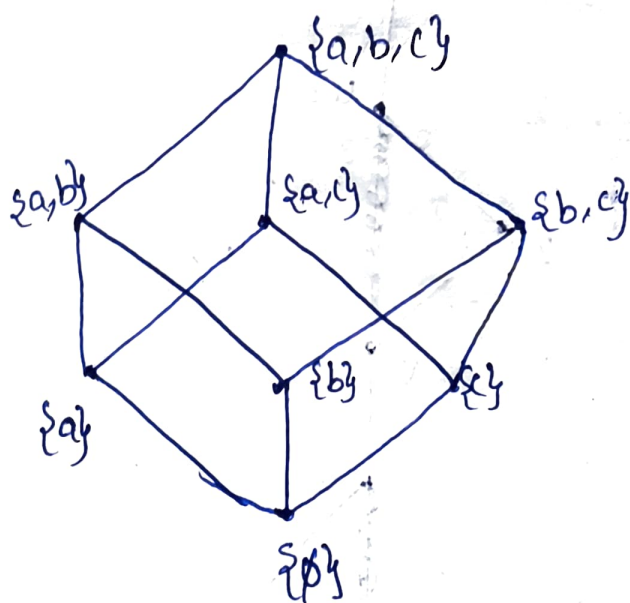
Given that

$B = \{a, b, c\}$ and the relation is ' $\subseteq$ '

Step 2:-

$$P(B) = \{\{\emptyset\}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\}\}$$

Step 3 :-



$$\text{infimum} = \{\emptyset\}$$

$$\text{supremum} = \{a, b, c\}$$

$$\text{no. of edges} = 12$$

b) Let $B = \{a, b, c, d\}$ and $A = P(B)$ p be power set of B. Draw the Hasse diagram for ' $\subseteq$ ' and poset A.

sol: step 1:

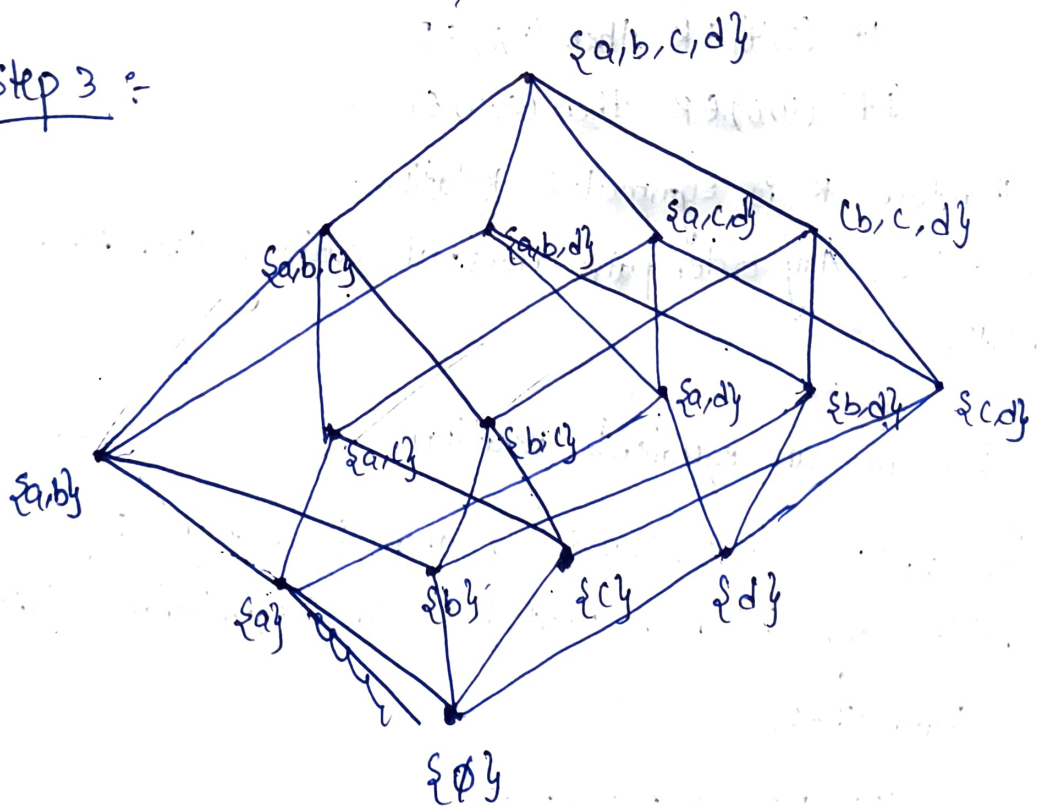
Given that

$B = \{a, b, c, d\}$ and the relation is ' $\subseteq$ '

step 2:

$$P(B) = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{b, c\}, \{c, d\}, \{d, a\}, \{a, c\}, \{b, d\}, \{a, b, c\}, \{b, c, d\}, \{c, d, a\}, \{b, a, d\}, \{a, b, c, d\} \}$$

step 3:



$$\text{infimum} = \{ \emptyset \}$$

$$\text{supremum} = \{ a, b, c, d \}$$

$$\text{no. of edges} = 32$$

Equality Relation:

1. If $S = \{1, 2, 3, \dots, n\}$ and a relation R on S where $R = \{ \langle x, y \rangle / x + y = 10 \}$. Prove that if R is an equivalent relation or not (equality $\rightarrow$ reflexive, symm, transitive)

Sol Given that,

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$R = \{(1, 9) (9, 1) (2, 8) (8, 2) (3, 7) (7, 3) (4, 6) (6, 4) (5, 5)\}$$

Reflexive:- only $(5, 5) \in R$ the remaining self order pairs does not exist in R . therefore R is not a Reflexive.

Symmetric:- If $(1, 9) \in R$ then $(9, 1) \in R$

If $(2, 8) \in R$ then $(8, 2) \in R$

If $(3, 7) \in R$ then $(7, 3) \in R$

If $(4, 6) \in R$ then $(6, 4) \in R$

Therefore R is Symmetric Relation

Transitive:- Any order pairs does not satisfy transitive relation

$\therefore R$ is not an equivalent Relation

- ★
2. Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and $R = \{ \langle x, y \rangle / (x-y) \text{ is divisible by } 3 \}$. Show that R is an equivalent relation.

Sol Given that,

$$X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$R = \{(1, 1) (2, 2) (3, 3) (4, 4) (5, 5) (6, 6) (7, 7) (8, 8) (9, 9) (1, 4) (4, 1) (1, 7) (7, 1) (2, 5) (5, 2) (2, 8) (8, 2) (3, 6) (6, 3) (3, 9) (9, 3) (4, 7) (7, 4) (5, 8) (8, 5) (6, 9) (9, 6)\}$$

reflexive: If R contains all self order pairs therefore R is Reflexive.

symmetric: If $(1,4) \in R$ then $(4,1) \in R$

If $(1,7) \in R$ then $(7,1) \in R$

If $(2,5) \in R$ then $(5,2) \in R$

If $(2,8) \in R$ then $(8,2) \in R$

If $(3,6) \in R$ then $(6,3) \in R$

If $(3,9) \in R$ then $(9,3) \in R$

If $(4,7) \in R$ then $(7,4) \in R$

If $(5,8) \in R$ then $(8,5) \in R$

If $(6,9) \in R$ then $(9,6) \in R$

$\therefore R$ is a symmetric

Transitive: If $(1,4) \in R$ and $(4,7) \in R$ then $(1,7) \in R$

If $(2,5) \in R$ and $(5,8) \in R$ then $(2,8) \in R$

If $(3,6) \in R$ and $(6,9) \in R$ then $(3,9) \in R$

If $(4,1) \in R$ and $(1,7) \in R$ then $(4,7) \in R$

If $(2,8) \in R$ and $(8,5) \in R$ then $(2,5) \in R$

If $(6,3) \in R$ and $(3,9) \in R$ then $(6,9) \in R$

If $(3,9) \in R$ and $(9,6) \in R$ then $(3,6) \in R$

$\therefore R$ is transitive

$\therefore R$ satisfies Reflexive, symmetric, Transitive

$\therefore R$ is an equivalence relation

3. Prove that the Relation "congruence Modulo 'm'" over the set of positive integers is an equivalence Relation

sol Let X be the set of positive integers and R is a relation congruence Modulo m defined as
(Congruence sym)

$$a \equiv b \pmod{m} \Leftrightarrow m \mid (a-b) \text{ (or) } \frac{a-b}{m}$$

Reflexive:-

Let $a, m \in X$

w.k.t

$$0 = \frac{a-a}{m}$$

$$a \equiv a \pmod{m} \quad \forall a \in X$$

' $\equiv$ ' is Reflexive

Symmetric:-

Let $a, b, m \in X$

$$\text{If } a \equiv b \pmod{m} \Leftrightarrow \frac{a-b}{m}$$

$$\Leftrightarrow -\frac{(a-b)}{m}$$

$$\Leftrightarrow \frac{b-a}{m}$$

$$\Leftrightarrow b \equiv a \pmod{m}$$

' $\equiv$ ' is symmetric

Transitive:-

Let $a, b, c, m \in X$

$$\text{If } a \equiv b \pmod{m} \Leftrightarrow \frac{a-b}{m}$$

$$\text{If } b \equiv c \pmod{m} \Leftrightarrow \frac{b-c}{m}$$

$$\Leftrightarrow \frac{a-b+b-c}{m}$$

$$\Leftrightarrow \frac{a-c}{m}$$

$$\Leftrightarrow a \equiv c \pmod{m}$$

' $\equiv$ ' is Transitive

$\therefore$ Congruence ($\equiv$) is satisfies Reflexive, symmetric, Transitive relation.

$\therefore$ congruence ($\equiv$) is an equivalence Relation

4) Let R denote a relation on the set of $\mathbb{R}^n$ order pairs of positive integers. Let it can be defined as $(x, y) R (u, v) \Leftrightarrow xv = yu$. show that R is an equivalence relation.

Let X is the set of positive integers
 R is the relation defined on X such that
 $(x, y) R (u, v) \Leftrightarrow xv = yu$.

Reflexive

Let $x, y \in X$

w.k.t $xy = yx$

$(x, y) R (x, y)$

$\therefore 'R'$ is Reflexive

Symmetric

Let $x, y, u, v \in X$

If $(x, y) R (u, v) \Leftrightarrow xv = yu$

$\Leftrightarrow yu = xv$

$\Leftrightarrow uy = vx$

$\Leftrightarrow (u, v) R (x, y)$

$\therefore 'R'$ is symmetric

Transitive

Let $x, y, u, v, m, n \in X$

If $(x, y) R (u, v) \Leftrightarrow xv = yu$

If $(u, v) R (m, n) \Leftrightarrow un = vm$

$\Leftrightarrow \cancel{xv} \cdot \cancel{un} = \cancel{yu} \cdot \cancel{vm} \Leftrightarrow x \cancel{v} \cancel{u} n = y \cancel{u} \cancel{v} m$

$\Leftrightarrow xn = ym$

$\Leftrightarrow (x, y) R (m, n)$

$\therefore 'R'$ is Transitive

$\therefore R$ is Reflexive, symmetric and transitive relation

$\therefore R$ is equivalence relation

Transitive closer

The relation obtained by adding the least no. of order pair to endure transitivity is called transitive closer of the relation or and denoted by R^* or R^+

$$R^* = R \cup R^2 \cup R^3 \cup \dots$$

1) Let $X = \{1, 2, 3, 4\}$ and $R = \{(1, 2) (2, 3) (3, 4)\}$ ⁱⁿ a relation on the set X what is the transitive closer of R

sol: Given that

$$X = \{1, 2, 3, 4\} \text{ and } R = \{(1, 2) (2, 3) (3, 4)\}$$

$$R^2 = R \overset{\text{circle}}{O} R$$

$$= \{(1, 2) (2, 3) (3, 4)\} O \{(1, 2) (2, 3) (3, 4)\}$$

$$= \{(1, 3) (2, 4)\}$$

$$R^3 = R^2 O R$$

$$= \{(1, 3) (2, 4)\} O \{(1, 2) (2, 3) (3, 4)\}$$

$$= \{(1, 4)\}$$

$$R^+ = R \cup R^2 \cup R^3$$

$$= \{(1, 2) (2, 3) (3, 4) (1, 3) (2, 4) (1, 4)\}$$

2) Let the relation R be $R = \{(1,2) (2,3) (3,3)\}$ on the set $A = \{1,2,3\}$ what is the Transitive closer of R
sol Given that

$$A = \{1,2,3\} \text{ and}$$

$$R = \{(1,2) (2,3) (3,3)\}$$

$$R^2 = R \circ R$$

$$= \{(1,2) (2,3) (3,3)\} \circ \{(1,2) (2,3) (3,3)\}$$

$$= \{(1,3) \}$$

$$R^3 = R^2 \circ R$$

$$= \{(1,3) (2,3)\} \circ \{(1,2) (2,3) (3,3)\}$$

$$R^+ = R \cup R^2$$

$$= \{(1,2) (2,3) (3,3) (1,3)\}$$

3) Define Partial ordering relation explain with an example

sol: ~~xxx~~

Partial ordering: A relation satisfies a reflexive, anti symmetric, Transitive then the relation is called partial ordering relation

ex:

Let X be the set of ^{subsets} positive integers and the relation is ' $\subseteq$ '

reflexive:- we know that $A \subseteq A \quad \forall A \in X$

$\therefore \subseteq$ is reflexive

anti symmetric:- Let A, B any two non empty sets

if $A \subseteq B$ and $B \subseteq A$ then $A = B$

$\therefore$ The relation ' $\subseteq$ ' is anti symmetric

Transitive:- Let A, B, C are any three non-empty sets

if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$

$\therefore$ The relation ' $\subseteq$ ' is transitive

$\therefore$ The relation ' $\subseteq$ ' is partial ordering Relation

Functions:

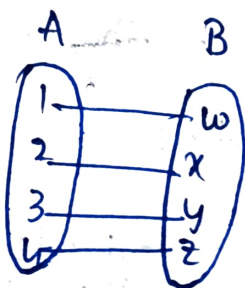
1) If $A = \{1, 2, 3, 4\}$, $B = \{w, x, y, z\}$ and if $f = \{(1, w) (2, x) (3, y) (4, z)\}$ then prove that f is both one-one and onto

sol Given that,

$$A = \{1, 2, 3, 4\}$$

$$B = \{w, x, y, z\}$$

$$f = \{(1, w) (2, x) (3, y) (4, z)\}$$



One-one: Every element in the domain of f have a unique image in the co-domain of f

$$\therefore f(1) = w$$

$$f(2) = x$$

$$f(3) = y$$

$$f(4) = z$$

$\therefore f$ is one-one

on-two: The number of elements in the codomain of $f = 4$,

Range of $f = 4$

$\therefore$ Codomain of $f =$ Range of f

$\therefore f$ is onto

$\therefore f$ is both one-one and on-two

$\therefore f$ is bijective function.

2) Let $X = \{1, 2, 3\}$ and f, g, h, s are functions from $X \rightarrow X$ given by $f = \{(1, 2) (2, 3) (3, 1)\}$, $g = \{(1, 2) (2, 1) (3, 3)\}$, $h = \{(1, 1) (2, 2) (3, 1)\}$, $s = \{(1, 1) (2, 2) (3, 3)\}$.
Find $f \circ g, g \circ f, f \circ h \circ g, s \circ g, g \circ s, s \circ s$ and $f \circ s$

sol Given that,

$$X = \{1, 2, 3\}$$

$$f = \{(1, 2) (2, 3) (3, 1)\}$$

$$g = \{(1, 2) (2, 1) (3, 3)\}$$

$$h = \{(1, 1) (2, 2) (3, 1)\}$$

$$s = \{(1, 1) (2, 2) (3, 3)\}$$

$$f \circ g = \{(1,2) (2,3) (3,1)\} \circ \{(1,2) (2,1) (3,3)\}$$

$$= \{(1,3) (2,2) (3,1)\}$$

$$f \circ h \circ g = \{(1,2) (2,3) (3,1)\} \circ \{(1,1) (2,2) (3,1)\} \circ \{(1,2) (2,1) (3,3)\}$$

$$= \{(1,2) (2,3) (3,1)\} \circ \{(1,2) (2,1) (3,1)\}$$

$$= \{(1,3) (2,2) (3,2)\}$$

$$g \circ f = \{(1,2) (2,1) (3,3)\} \circ \{(1,2) (2,3) (3,1)\}$$

$$= \{(1,1) (2,3) (3,2)\}$$

$$s \circ g = \{(1,1) (2,2) (3,3)\} \circ \{(1,2) (2,1) (3,3)\}$$

$$= \{(1,2) (2,1) (3,3)\}$$

$$g \circ s = \{(1,2) (2,1) (3,3)\} \circ \{(1,1) (2,2) (3,3)\}$$

$$= \{(1,2) (2,1) (3,3)\}$$

$$s \circ s = \{(1,1) (2,2) (3,3)\} \circ \{(1,1) (2,2) (3,3)\}$$

$$= \{(1,1) (2,2) (3,3)\}$$

$$f \circ s = \{(1,2) (2,3) (3,1)\} \circ \{(1,1) (2,2) (3,3)\}$$

$$= \{(1,2) (2,3) (3,1)\}$$

3) Let $f(x) = x+2$, $g(x) = x-2$ and $h(x) = 3x \forall x \in \mathbb{R}$ where $\mathbb{R}$ is the set of real numbers. Find $g \circ f$, $f \circ g$, $f \circ f$, $g \circ g$, $f \circ h$, $h \circ g$, $h \circ f$ and $f \circ h \circ g$.

Sol Given that,

$$f(x) = x+2$$

$$g(x) = x-2$$

$$h(x) = 3x$$

$$i) g \circ f(x) = g[f(x)]$$

$$= (x+2)-2$$

$$= x$$

$$ii) f \circ g(x) = f[g(x)]$$

$$= f[x-2]$$

$$= (x-2)+2$$

$$= x$$

$$iii) f \circ f(x) = f[f(x)]$$

$$= f(x+2)$$

$$= (x+2)+2$$

$$= x+4$$

$$iv) g \circ g(x) = g[g(x)]$$

$$= g(x-2)$$

$$= (x-2)-2$$

$$= x-4$$

$$\begin{aligned} \text{v) } f \circ h(x) &= f(h(x)) \\ &= f(3x) \\ &= 3x+2 \end{aligned}$$

$$\begin{aligned} \text{vi) } h \circ g(x) &= h(g(x)) \\ &= h(x-2) \\ &= 3(x-2) \\ &= 3x-6 \end{aligned}$$

$$\begin{aligned} \text{vii) } h \circ f(x) &= h(f(x)) \\ &= h(x+2) \\ &= 3(x+2) \\ &= 3x+6 \end{aligned}$$

$$\begin{aligned} \text{(viii) } f \circ h \circ g(x) &= f(h(g(x))) \\ &= f(h(x-2)) \\ &= f(3(x-2)) \\ &= f(3x-6) \\ &= (3x-6)+2 \\ &= 3x-4 \end{aligned}$$

4) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^3 - 2$ then find

f 's inverse

sol Given that

$$f(x) = x^3 - 2$$

f ' - one-one

$$f(x) = x^3 - 2$$

$$f(x_1) = x_1^3 - 2$$

$$f(x_2) = x_2^3 - 2$$

$$f(x_1) = f(x_2) \Rightarrow x_1^3 - 2 = x_2^3 - 2$$

$$x_1^3 = x_2^3$$

$$x_1 = x_2$$

f ' is one-one

f ' - onto

$$\text{Let } y = f(x) = x^3 - 2$$

$$y = x^3 - 2$$

$$x^3 = y + 2$$

$$x = (y + 2)^{1/3}$$

$$f(x) = x^3 - 2$$

$$f((y+2)^{1/3}) = ((y+2)^{1/3})^3 - 2$$

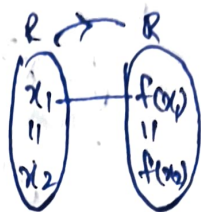
$$= y + 2 - 2$$

$$= y$$

f is onto

If f is both one-one and onto

$\therefore f$ is bijective function



find the f^{-1} inverse

$$\text{Let } y = f(x) = x^3 - 2$$

$$y = x^3 - 2$$

$$x^3 = y + 2$$

$$x = (y + 2)^{1/3}$$

$$y = f(x)$$

$$x = f^{-1}(y)$$

$$(y + 2)^{1/3} = f^{-1}(y)$$

$$f^{-1}(x) = (x + 2)^{1/3}$$

5) show that $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x - 3$ is a bijection and find its inverse. compute $f \circ f$ and $f \circ f^{-1}$

sol Given that

$$f(x) = 2x - 3$$

f' is one-one

$$f(x) = 2x - 3$$

$$f(x_1) = 2x_1 - 3$$

$$f(x_2) = 2x_2 - 3$$

$$f(x_1) = f(x_2) \Rightarrow 2x_1 - 3 = 2x_2 - 3$$

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

f' is one-one

f' is onto

$$\text{Let } y = f(x) = 2x - 3$$

$$y = 2x - 3$$

$$2x = y + 3$$

$$x = \frac{y+3}{2}$$

$$f(x) = 2x - 3$$

$$f\left(\frac{y+3}{2}\right) = 2\left(\frac{y+3}{2}\right) - 3$$

$$= y + 3 - 3$$

$$= y$$

"f" is onto

find the f^{-1} inverse

$$\text{let } y = f(x) = 2x - 3$$

$$y = 2x - 3$$

$$2x = y + 3$$

$$x = \frac{y+3}{2}$$

$$y = f(x)$$

$$f^{-1}(y) = x$$

$$\frac{y+3}{2} = f^{-1}(y)$$

$$f^{-1}(x) = \frac{x+3}{2}$$

$$(f \circ f^{-1})(x) = f[f^{-1}(x)]$$

$$= f\left[\frac{x+3}{2}\right]$$

$$= \frac{2\left(\frac{x+3}{2}\right) - 3}{2}$$

$$= x$$

$$f \circ f^{-1}(x) = f(f^{-1}(x))$$

$$= f\left(\frac{x+3}{2}\right)$$

$$= 2\left(\frac{x+3}{2}\right) - 3 = x + 3 - 3 = x$$

c) Define Lattice and write the properties

2M

Lattice:- A Lattice L is a special kind of Partially ordered set in which every pair of elements in L have least upper bound ($a \vee b$) and greater lower bound ($a \wedge b$) $\forall a, b \in L$.

Properties:

i) idempotent:

$$a \wedge a = a$$

$$a \vee a = a$$

$$\forall a \in L$$

ii) Commutative:

$$a \vee b = b \vee a$$

$$\forall a, b \in L$$

$$a \wedge b = b \wedge a$$

iii) associative:

$$a \vee (b \vee c) = (a \vee b) \vee c$$

$$a \wedge (b \wedge c) = (a \wedge b) \wedge c$$

$$\forall a, b, c \in L$$

iv) absorption:

$$a \vee (a \wedge b) = a$$

$$a \wedge (a \vee b) = a$$

$$\forall a, b \in L$$

v) Complement:

$$a \vee a' = 1$$

$$a \wedge a' = 0$$

$$\forall a, a' \in L$$

vi) distributive:

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$\forall a, b, c \in L$$

Unit - 3

Permutations and Combinations and Recurrence Relations

Part - A (Permutations and Combinations)

1. Introduction to Permutations
2. Permutations without repetitions
3. Permutations with repetitions
4. Introduction to Combinations
5. Combinations without repetitions
6. Integral equations
7. Binomial theorem
8. Multinomial Theorem

Part - B (Recurrence Relations)

Permutations

$$1) {}^n P_r \text{ (or) } P(n, r) = \frac{n!}{(n-r)!}$$

$$2) 0! = 1$$

$$3) {}^n P_0 = 1$$

$$4) {}^n P_1 = n$$

$$5) {}^n P_n = n!$$

$$6) n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1$$

DOM
v.vimp
1) Find n if

$$(i) P(n, 2) = 72$$

$$(ii) P(n, 4) = 42 P(n, 2)$$

$$(iii) 2P(n, 2) + 50 = P(2n, 2)$$

(ii) Given that

$$P(n, 4) = 42 P(n, 2)$$

$$\frac{n!}{(n-4)!} = 42 \frac{n!}{(n-2)!}$$

$$\frac{1}{(n-4)!} = 42 \frac{1}{(n-2)!}$$

$$\frac{1}{\cancel{(n-4)!}} = 42 \frac{1}{(n-2)(n-3)\cancel{(n-4)!}}$$

$$1 = 42 \frac{1}{(n-2)(n-3)}$$

$$(n-2)(n-3) = 42$$

$$n^2 - 5n + 6 = 42$$

$$n^2 - 5n - 36 = 0$$

$$n^2 - 9n + 4n - 36 = 0$$

$$n(n-9) + 4(n-9) = 0$$

$$(n-9)(n+4) = 0$$

$$n = 9, -4$$

$$\boxed{n=9}$$

$$(iii) \quad 2P(n, 2) + 50 = P(2n, 2)$$

$$2 \frac{n!}{(n-2)!} + 50 = \frac{(2n)!}{(2n-2)!}$$

$$2 \frac{n(n-1)(\cancel{n-2})!}{(\cancel{n-2})!} + 50 = \frac{2n(2n-1)(\cancel{2n-2})!}{(\cancel{2n-2})!}$$

$$n^2 - n + 25 = 2n^2 - n$$

$$n^2 = 25$$

$$n = \pm 5$$

$$\boxed{n=5}$$

(i) Given that

$$P(n, 2) = 72$$

$$\frac{n!}{(n-2)!} = 72$$

$$\frac{n(n-1)(n-2)!}{(n-2)!} = 72$$

$$n^2 - n = 72$$

$$n^2 - n - 72 = 0$$

$$n = 9, -8$$

$$\boxed{n = 9}$$

Permutations without Repeats

1) How many possibilities are there to select a first prize winner, a second prize winner and a third prize winner from 50 different people who have entered a contest.

Sol: Total no. of people = 50
Total no. of prizes = 3

The total no. of possibilities to select a first prize, second prize, third prize winners from 50 people

$$= {}_{50}P_1 \times {}_{49}P_1 \times {}_{48}P_1$$

$$= 50 \times 49 \times 48$$

$$= 117600$$

2) Consider the 6 digits 1, 2, 3, 5, 6, 7 assuming that repetitions are non permitted answer the following

- i) How many 4 digit numbers can be formed from the 6 digits 1, 2, 3, 5, 6, 7
- ii) How many of these numbers are less than 4000
- iii) How many of the numbers are even
- iv) How many of the numbers are odd
- v) How many of the numbers are multiples of 5
- vi) How many of the numbers contains both the digits 5 and 7

Sol:-

(i) The total number of 4 digit numbers can be formed using the digits 1, 2, 3, 5, 6, 7 = 6P_4
= 360 members

(ii) The total number of 4 digit number which are less than 4000 = ${}^3P_1 \times {}^5P_3$
$$= 3 \times \frac{5 \times 4 \times 3 \times 2}{2}$$

= 180 members

(iii) The total number of even numbers = ${}^5P_3 \times {}^2P_1$
$$= 5 \times 4 \times 3 \times 2$$

= 120 members

(iv) The total no. of odd numbers = $360 - 120$
= 240 members

(v) The total no. of 4 digit numbers which are multiples of 5 = ${}^5P_3 \times {}^1P_1 = 20 \times 1 = 20$ members

vi) The total no. of 4 digit numbers which contains both 5 and 7 = ${}^4P_2 \times {}^4P_2$

$$= 12 \times 12$$

$$= 144 \text{ members}$$

3) Suppose repetitions are not permitted answer the following questions

i) How many 3 digit numbers can be formed from the 6 digits 2, 3, 5, 6, 7, 9 [6P_3]

ii) How many of these numbers are less than 400 (${}^2P_1 \times {}^5P_2$)

iii) How many of these are even (${}^5P_2 \times {}^2P_1$)

~~iv) How many permutations of the letters A, B, C, D, E,~~

4) How many of the letters A, B, C, D, E, F, G, H contains.

- i) The string CDE
- ii) The strings AB & FG
- (iii) The strings ABC & CDE
- iv) The string ABC and BED

Sol

(i) The no. of permutations of the string CDE
 $= 3! = 6$

(ii) The no. of permutations of the strings AB and FG
 $= 2! \times 2! = 4$

(iii) The no. of permutations of the strings ABC and CDE
 $= 3! \times 3! = 36$

(iv) The no. of permutations of the strings ABC and BED
 $= 3! \times 3! = 36$

5) In how many ways can 4 mathematics books, 3 history books, 3 chemistry books and 2 social books be arranged on a shelf so that all the books of the same subject are together

Sol: The total no. of ways can be arranged on a shelf all the books of the same subject together

$$= 4! \times 3! \times 3! \times 2!$$

$$= 24 \times 24 \times 6 \times 6 \times 2$$

$$= 41472$$

b) i) In how many ways can 6 men and 4 women sit in a row $(10!)$

ii) In how many ways can they sit in a row if all the men sit together and all the women sit together $(2! \times 6! \times 4!)$

iii) In how many ways can they sit in a row if just the women sit together $(7! \times 4!)$

iv) In how many ways can they sit in a row if just the men sit together $(5! \times 6!)$

4) In how many different orders can three men, three women be seated in a row of 6 seats if

i) Any one may sit in any of the seats $(6!)$

ii) The first and last seats must be filled by men

$(3P_2 \times 4!)$

~~iii) In how many ways can they sit in a row if~~

iii) Men occupy the first three seats last 3 seats $(3! \times 3!)$

sol

Permutations with Repetitions

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

1) How many different strings can be made from the letters of the word "SUCCESS" using all the letters

sol The total number of letters = 7

Number of letters S's = 3

Number of letters U's = 1

Number of letter C's = 2

Number of letter E's = 1

The total no. of strings can be made from the letters of the word success = $\frac{7!}{3! \times 1! \times 2! \times 1!}$

$$= 420$$

2) How many different strings can be made from the letters of the word "ABRACADABRA" using all the letters.

Sol

The total no. of letters = 11

no. of letter A's = 5

no. of letter B's = 2

no. of letter R's = 2

no. of letter C's = 1

no. of letter D's = 1

The total no. of strings can be made from the letters of the word ABRACADABRA

$$= \frac{11!}{5! \times 2! \times 2! \times 1! \times 1!}$$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5! \times 2! \times 2!}$$

~~207360~~

$$= 83160$$

Binomial Theorem:

If x, y are any two real numbers and for $n > 0$ then

$$(x+y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y^1 + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_n y^n$$
$$= \sum_{i=0}^n {}^nC_i x^{n-i} y^i$$

1) Find the coefficient of $x^{101} y^{99}$ in the expansion of $(2x-3y)^{200}$

sol:-

The coefficient of $x^{101} y^{99}$ in the expansion of $(2x-3y)^{200}$ is

$$\frac{200!}{101! 99!} \cdot 2^{101} (-3)^{99}$$

2) Find the coefficient of $x^9 y^3$ in the expansion of $(2x-3y)^{12}$

sol

The coefficient of $x^9 y^3$ in the expansion of $(2x-3y)^{12}$ is

$$\frac{12!}{9! 3!} \times 2^9 \times (-3)^3$$

3) Find the coefficient of $x^5 y^8$ in the expansion of $(x+y)^{13}$

sol

The coefficient of $x^5 y^8$ in the expansion of $(x+y)^{13}$ is

$$\frac{{}^{13}C_5}{5! \times 8!}$$

Multinomial Theorem:

If $a_1, a_2, \dots, a_k$ are any real numbers and for $n > 0$ then

$$(a_1 + a_2 + \dots + a_k)^n = \sum_{r_1 + r_2 + \dots + r_k = n} \frac{n!}{r_1! r_2! \dots r_k!} a_1^{r_1} a_2^{r_2} \dots a_k^{r_k}$$

1) Expand the multinomial $(x_1 + x_2 + x_3 + x_4)^4$

$$(x_1 + x_2 + x_3 + x_4)^4 = \sum_{r_1 + r_2 + r_3 + r_4 = 4} \frac{4!}{r_1! r_2! r_3! r_4!} x_1^{r_1} x_2^{r_2} x_3^{r_3} x_4^{r_4}$$

2) What is the coefficient of $v^2 w^3 x^4 y^2$ in the expansion of $(v + v + 2w + x + y + z)^{11}$

Sol: The coefficient of $v^2 w^3 x^4 y^2$ in the expansion

$$(v + v + 2w + x + y + z)^{11} = \frac{11!}{2! 3! 4! 2!} 1^2 x^2 2^3 x^4 x^2$$

3) Find the coefficient of $x^2 y^4 z^4$ in the expansion of $(2x - y + 3z)^{10}$.

Sol: The coefficient of $x^2 y^4 z^4$ in the expansion of

$$(2x - y + 3z)^{10} = \frac{10!}{2! 4! 4!} 2^2 x^2 (-1)^4 x (3)^4$$